

Frontend Pipeline

Autonomous Multi-Agent Website Redesign Platform

From audit to production-ready Next.js in one command

The Problem

Traditional website redesign is slow, manual, and error-prone:

- Manual audits & research
- Art direction & design
- Multiple specialist handoffs
- Revision cycles measured in days
- Weeks to months for delivery

4-12 weeks for traditional redesign

The Solution

10 specialized AI agents coordinated by a Master Orchestrator

15-30 minutes for complete delivery

- Parallel execution where possible
- Typed contracts between agents
- Automated QA & rework loops
- Production-ready Next.js output

10 Core Capabilities

1. Website Analysis - Full-site audit with screenshots
2. SEO Audit - Lighthouse + Core Web Vitals
3. Brand Research - Brand DNA + competitors
4. Lead Discovery - Top 20 prospects with scoring
5. Creative Direction - Art direction + moodboards
6. UX Planning - Sitemap + wireframes
7. UI Design - Design tokens + components
8. Motion Design - Animation specs
9. Frontend Engineering - Production Next.js code
10. QA Gate - 7-dimension quality score

Pipeline Architecture

5 Sequential Phases

Phase 1: Analysis (Parallel)

3 agents run simultaneously:

- **Website Analysis** - Crawl, screenshots, vision audit
- **SEO** - Lighthouse, Core Web Vitals, accessibility
- **Brand Research** - Brand DNA, competitors, audience

Phase 2: Creative Direction

Single source of truth for the redesign:

- Color System
- Typography System
- Moodboard (4-6 references)
- Signature Moment (unique interaction)
- Design Principles

Phase 3: Experience Design (Parallel)

- **UX** - Sitemap, wireframes, IA, CTA hierarchy
- **UI Design** - Tokens, components, spacing, responsive

Phase 4: Engineering & QA

- **Engineering** - Next.js 15 with full tech stack
- **QA** - 7-dimension quality gate

Phase 5: Delivery

Complete package including:

- Production-ready Next.js project
- 10 comprehensive reports
- Deploy-ready zip file
- QA score & file manifest

7-Dimension QA Gate

Responsiveness	Mobile, tablet, desktop
Lighthouse	Performance, accessibility, SEO
Accessibility	ARIA, contrast, keyboard nav
SEO	Meta tags, headings, schema
Performance	Bundle size, lazy loading
Animation	Scroll narrative, easing
Consistency	Token adherence, reuse

Generated Tech Stack

- **Framework:** Next.js 15 (App Router)
- **Language:** TypeScript
- **Styling:** Tailwind CSS 4
- **Animation:** GSAP, Framer Motion, Lenis
- **3D:** Three.js + @react-three/fiber
- **Fonts:** Google Fonts
- **Icons:** Custom SVG system

Cost Analysis

Negligible API costs, massive margin opportunity

Per-Website Cost

LLM (Primary)	\$0.40–\$0.80
LLM (Secondary)	\$0.05–\$0.10
LLM (Creative)	\$0.15–\$0.30
Infrastructure	\$12/mo shared

Total: \$0.60–\$1.30 per website

Revenue Model

Service	Price	Cost	Margin
Website Audit	\$2-5K	\$1	99.9%
Redesign Proposal	\$10-25K	\$1.50	99.9%
Full Redesign	\$25-75K	\$1.50	99.9%

Lead Discovery & Scoring

Automatic ranking of top 20 prospects by:

SEO Score	25%
Design Score	25%
Lead Score	25%
Pain Points	25%

Each prospect includes identified weaknesses + personalized redesign pitch

Style Mode System

Prevent template convergence with committed design directions:

- **Editorial:** Serif typography, asymmetric grid, measured motion
- **Brutalist:** Monospace accents, high contrast, snappy motion
- **Ambient:** Low contrast, restrained type, slow parallax
- **Kinetic Type:** Typography as hero, scroll-scrubbed transforms

Competitive Advantages

SPEED

15-30 min vs 4-12 weeks

CONSISTENCY

7-dim QA gate per delivery

DATA-DRIVEN

SEO/design scores, not guesses

SCALABILITY

10 redesigns in parallel

Reports Delivered

- Website Analysis - Full UI/UX audit
- SEO & Performance - Lighthouse scores + plan
- Brand DNA - Brand positioning document
- Lead Report - Top 20 prospects with pitches
- Creative Direction - Art direction + moodboards
- UX Plan - Sitemap + wireframes
- UI System - Design tokens + components
- Motion Specs - Animation specifications
- Engineering Guide - Build instructions + code summary
- QA Report - Quality gate scores

Quick Start

Minimal requirements and one-command setup

Requirements

- Python 3.10+
- Node.js 18+
- PostgreSQL (local or managed)
- Anthropic API key

One-Command Execution

Risk Mitigation

- **Quality Variance:** Mandated minimum models; escalation to Opus
- **Generic Output:** Style mode system + Generic SaaS detection
- **Build Failures:** Internal npm build validation
- **Prompt Injection:** Schema validation on all inputs
- **Security:** Restricted crawling, timeouts, size limits

Service Tiers

Tier	Price	Turnaround
Audit	\$2-5K	1 hour
Proposal	\$10-25K	2-4 hours
Full Redesign	\$25-75K	4-8 hours
Retainer	\$2-5K/mo	Ongoing

Summary

The Frontend Pipeline transforms website redesign from a 4-12 week manual process into a 15-30 minute autonomous operation.

- **Faster:** 10x speed improvement
- **Consistent:** Every project passes 7-dimension QA
- **Data-Driven:** Measurable SEO and design scores
- **Profitable:** 99.9% margin per delivery

Questions?

Frontend Pipeline - Autonomous Multi-Agent Website Redesign